
**Guideline for dyeing of cotton fabrics in
batch, semi continuous and continuous
process with reactive dyes**

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Foreword

This Ethiopian Standard has been prepared under the direction of Technical Committee for Textile-Dyeing and finishing (TC 71) and published by the Ethiopian Standards Agency (ESA).

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DRAFT STANDARD

Guideline for Dyeing of Cotton Fabrics in Batch, Semi Continuous and Continuous Process with Reactive Dyes

1. SCOPE:

This Ethiopian standard specifies the guideline for dyeing of cotton fabrics in batch, semi continuous and continuous process with reactive dyes. This standard does not specify dyeing cotton with vat dyes, sulphur dyes, direct dyes, azoic dyes and indigo dyes.

2. NORMATIVE REFERENCE

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ES 6265: Terminology Relating to Textiles

ES 6702: Sustainable textile production

ES 6736 Cleaner production for Textile Industries.

ES 6583 Code of practice for grading of textile fabrics

3. TERMS AND DEFINITION

For the purpose of this standard the following definitions shall apply:-

3.1.

Dyeing

A method for colouring a textile material in which a dye is applied to the substrate in a uniform manner to obtain an even shade with a performance and fastness appropriate to its final use.

3.2.

Dyestuff

A molecule which contains a chromophoric group capable of interacting with light, thus giving the impression of colour.

3.3.

Reactive Dyes

They are anionic in nature, water soluble they have better substantivity. They form strong covalent bond with cellulosic fibers. The main feature of the dyestuff is its low affinity to cellulose; therefore large amounts of salt are required to force its deposition on the fabric. They have better wash and light fastness. Alkaline condition must be required for dyeing.

3.4.

Substantivity

Tendency of a dye to move from a solution out of fibres in the dye solution. It is a quality of dye. Without substantivity, most of dyes would simply remain in solution or dispersion in the bath.

3.5.

Exhaustion

The process of transfer of dyestuff from the dye bath on to the fibre or material is known as exhaustion.

3.6.

Fixation

Formation of the "final" bond between the dye and the fibre with the mechanisms such as ionic bonding and hydrophobic forces.

3.7.

Compatibility

A state in which two things are able to exist or occur together without problems or conflict. Normally a combination of several dyestuffs is necessary to obtain a given colour. The behavior of dyes in admixture is therefore of great practical interest. Level and repeatable dyeings are achieved more easily if dyes of almost equal behavior in admixture can be selected.

3.8.

Depth of shade

It is a ratio of weight of dye to weight of goods dyed, usually expressed as percentage; amount of dye, owf (over the weight of fabric)

3.9.

Color fastness

The resistance ability of dye molecule to any agency named as washing, light, rubbing and crocking.

3.10.

Level Dyeing

Uniform in shade over the surface of a piece of dyed fabric or along the length of dyed yarn Level dyeing is usually the objective in commercial processes.

3.11.

Padding

This is the most important component of semi or continuous dyeing machines. It is a dyeing method with very low liquor to goods ratio, where typically only enough strong dye solution is used to saturate the fabric to fix the dye quickly.

3.12.

Pick up%

It is a ratio of amount pick or uptake by fabric when it passes through the solution or dye liquor.

3.13.**Liquor Ratio**

It is the ratio of the weight of the dye bath or other processing bath to the weight of the goods being dyed or processed for immersion dyeing in art dyeing processes, common liquor to goods ratio is 20:1. That is, for each kilogram of fibre to be dyed, 20 kilograms of dye bath is used of the dye solution is to be absorbed by the fibre, such as padding.

3.14.**Tailing**

Tailing is refers to a phenomenon in which depth & shade occur on dyed fabric owing to worst distribution of dyes from top to bottom (first meters and last meters). This is a problem of semi & continuous dyeing.

3.15.**Listing**

Listing refers to a phenomenon in which variations in depth & shade occur on dyed fabric owing to uneven dye uptake from side to side and side to center. For preventing listing during padding, padding liquor should be taken up uniformly onto textile substrate from side to center, while the textile substrate are immersed in padding liquor for a very short time & then squeezed.

4. GENERAL PRECAUTIONS TO OBTAIN LEVEL DYEING WITH CONSISTENCY**4.1. Ready for dyeing (RFD) fabric**

- 4.1.1.** Prior to preparation, greige goods should be thoroughly inspected as ES 6583
- 4.1.2.** Fabric should have uniform absorbency and whiteness throughout the length and across.
- 4.1.3.** Residual size must always be removed from woven goods because of the risk of dye wastage by reaction with hydroxyl groups in size components.
- 4.1.4.** The fabrics should be free from alkali and should be in neutral condition
- 4.1.5.** Proper neutralization of core alkali prior to drying is essential to avoid shade change and hydrolysis of dye-fibre bond.
- 4.1.6.** Where bleaching is necessary, the bleached fabric should be free from traces of residual chlorine or peroxy compounds.
- 4.1.7.** There is considerable variation in the ability of reactive dyes to cover dead or immature cotton. For this reason it should be to causticize or mercerize woven fabrics in order to achieve a satisfactory appearance in certain hues.
- 4.1.8.** The fabric should also be free from earth metals & salts (Fe, Ca, Mg)
- 4.1.9.** The woven fabric pieces should be sown together tail-to-head, forming a sort of "ribbon". Fabric stitching should not be overlapped.
- 4.1.10.** The seam must be as flat as possible to avoid forming an impression on the fabric layers.
- 4.1.11.** Over drying, or drying too quickly at too high a temperature, may decrease dye penetration. Fabric should be cooled before rolling or plaiting.

4.1.12. Before padding the woven fabric should be properly conditioned and cooled so that moisture content and temperature of fabric is same throughout the batch.

4.1.13. It is essential that all the fabrics in the roll should be of the same width.

4.1.14. The fabric should be on roller with coincident selvages, i. e, no part of selvages should come out of the roll, as that portion will be dyed differently

4.2. Handling of reactive dyes

4.2.1. Because most reactive dyes are prone to hydrolysis, their handling and use should be with care.

4.2.2. Most are readily water-soluble and the dye solution should be prepared in the usual way by pasting with water and then adding more water.

4.2.3. Hot water should not be used for dissolving dyes of high reactivity, because of the risk of hydrolysis of the reactive group, but is suitable for the less reactive types.

4.2.4. Once the dye solution has been prepared, it should not be stored for later use without some risk of hydrolysis of the reactive group.

4.2.5. Vinyl sulphone (Dyes containing a 2-sulphatoethylsulphone group) however, can be dissolved in neutral water at the boil without risk of hydrolysis.

4.2.6. Reactive dyes from different manufacturers' ranges should not be used in the same recipe, unless it is known that they contain the same reactive system.

4.2.7. Dyestuffs used in combination should be compatible and have similar exhaustion and fixation profile.

4.2.8. Dyes should have good solubility in presence of salt and alkali at given dyeing temperature. Use of leveling agent, dye bath conditioner, solubilizing agent is recommended as preventive measure.

4.2.9. Pre-dissolved and filtered dyestuff solution is dosed linearly in the dyeing machine.

4.2.10. Liquor ratio for dyeing is adjusted in such a way that after adding dyestuff solution, salt solution and alkali solution, it is maintained at required ratio.

4.3. Selection of reactive dyes

4.3.1. The guiding principle governing dye selection should be to obtain the desired hue and colour fastness by an economical process that results in a good-quality dyed fabric.

4.3.2. Method of dyeing used should be as per recommendations by dyestuff supplier i.e. whether salt in beginning, dyestuff in beginning or all in one or isothermal dyeing method. For comparison of application properties of reactive dyes see the table 1 below:-

Table 1: comparison of application properties of reactive dyes

Property Dyes	Substantivity	Dyeing T°C	Fixation %	Wash ability	Stability to alkali	Stability to acid	Dyeing method		
							Batch	Continuous	Cold pad batch
VS	Low	50-60	60-70	Good	Bad	Good	Good	Good	Good
Supra	Medium	60-70	80-90	Good	Good	Good	Good	Good	Good
MCT	High	80	70-80	Bad	Good	Medium	Good	Good	Bad

VS--Vinyl sulphone

SUPRA Hetro bifunctional that carry monochlorotriazine and sulphato ethyl sulphone reactive systems

MCT-- Monochlorotriazines

- ✚ Factors affecting dyeing properties of reactive dyes: type and number of reactive group, chemical structure of chromogens, bridging group
- ✚ Factors affecting substantivity in reactive dyeing: Chemical structure of reactive dye, concentration of dye, liquor ratio, amount of neutral salt, dyeing temperature, rate of heating, pH, etc.

4.4. Dyeing process of reactive dyes

4.4.1. Dyeing process of reactive dyes for MCT (Monochlorotriazines), VS (Vinyl sulphone) and Supra (Hetro bifunctional that carry monochlorotriazine and sulphato ethyl sulphone reactive systems) should be as in the diagram shown below:-

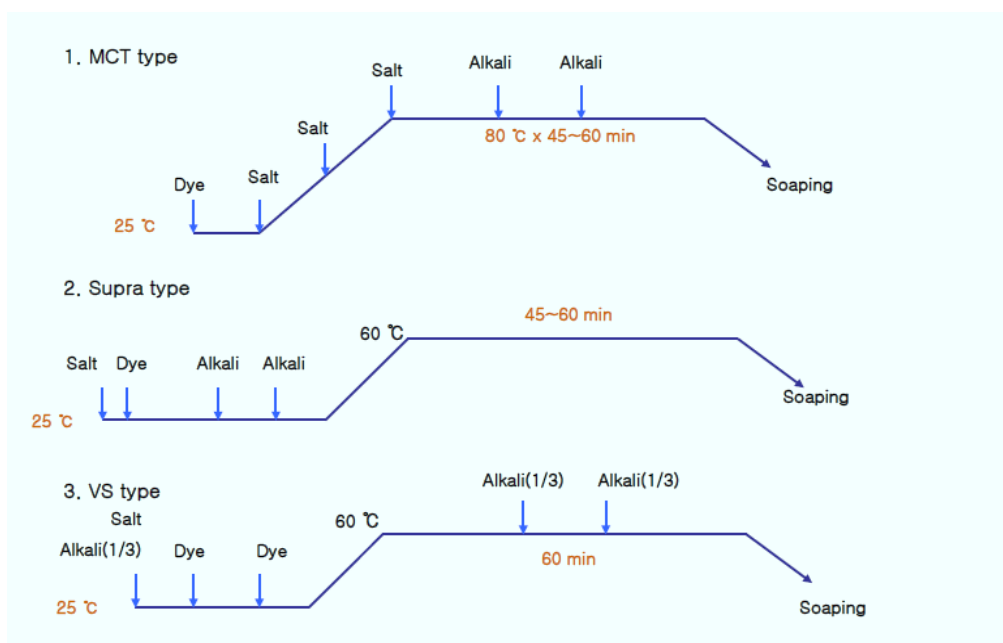


Diagram: Dyeing process of reactive dyes

4.5. Fastness properties of reactive dyes on cotton

4.5.1. The washing procedure should remove completely the unfixed and hydrolyzed dye. Apparent inferior fastness to washing is usually because of incomplete removal of unreacted and hydrolyzed dye from the material by washing after dyeing.

4.5.2. In particular, the fastness to bleaching by chlorine, and to a lesser extent by peroxides present in modern household detergents, is often only moderate. Therefore, care should be taken.

4.5.3. Any shade change due to dye fixing or finishing should be taken into account while finalizing the recipe of lab dips or submitting shade for approval.

4.6. Water quality for the dye house

4.6.1. Although seldom sensitive to neutral hard water, precipitation of hardness constituents results at the alkaline fixation stage and thus soft water should be used for all dissolving and dye bath operations. (Annex B)

4.6.2. Typical dye house water quality should be as table 2 below

Table 2: Typical dye house water quality

Component	Values
pH value	7.0–7.5
Total hardness (ppm, CaCO ₃)	10–25
Iron ((ppm)	0.02–0.1
Manganese (ppm)	0.03
Alkalinity to Methyl Orange (ppm, CaCO ₃)	35–65
Dissolved solids Total dissolved solids (ppm)	65–150
Suspended solids (ppm)	Nil
Chloride (ppm)	0–30
Sulphate (ppm)	0–30

5. BATCH DYEING PRINCIPLES, STANDARD OPERATING PROCEDURE AND PROCESS CONTROL

5.1. General principle of batch dyeing

A typical exhaust dyeing process for cellulosic materials using reactive dyes should follow the following three phases:

- The initial exhaustion phase
- The fixation phase
- The post-dyeing washing.

5.2. Standard operating procedure of batch dyeing

5.2.1. Dyeing of knitted fabric dyeing in winch dyeing machine

Knitted fabric dyeing by exhaust process in winch dyeing machines for hot brand reactive dyes should be as follows:-

(For bleached mercerized cotton using reactive dyes, liquor ratio 20:1, 3% C.I. Reactive dye 3%)

- Set the bath at room temperature-25°C
- Add the prepared dye solution
- Raise the temperature to 49°C and hold for 40 minute
- Add salt 100 gram/litre in four portions over 40 minute while raising the temperature to 80°C
- Dye at 80°C for 20 minute
- Then add 20 gram/litre soda ash and 1 gram/litre caustic soda
- Run at 80°C for 50 minute

- Drop bath
- Check the shade and adjust, if necessary
- Cold rinse 10 minute
- Drop bath
- Hot rinse at 65°C 10 minute
- Drop bath
- Set soaping bath at 38°C with 1 gram/litre anionic surfactant and 1 gram/litre soda ash, Run for 15 minute at 100°C
- Cool to 70°C
- Drop bath
- Give hot rinse 65°C 10 minute
- Give cold rinse 10 minute
- Drop bath and finish

5.3 Salt and alkali for hot brand reactive dyes should be as in the table 3 below:-

Salt and alkali concentration Shade %	Common salt or Glauber's salt (g/l)	Soda ash (g/l)	NaOH (g/l)
Below 0.5%	30	20	2
0.5% - 2.0%	45	20	2
2.0% - 4.0%	70	20	2
4.0% & Above	90	20	2

For shade adjustment, if dye addition is above 25% of the original recipe, a fresh bath should be set. If dye addition is below 25% of the original recipe half bath method or full bath method should be used.

Half bath shading

- Drop half of the bath and add cold water to reduce temperature and electrolyte concentration.
- Add well diluted pre-dissolved dye and dye for 10 minute.
- Raise the temperature to 80°C and dye for 30 minute 80°C.
- Check the shade.

The full bath is applicable when faster cooling is possible. The method is

- Cool dye bath to 60°C
- Add well diluted pre-dissolved dye and dye for 10 minute
- Raise the temperature to 80°C and dye for 30 minute at 80°C
- Check the shade

5.2.2. Dyeing of woven fabrics in jigger dyeing machine (for cold brand reactive dyes)**5.2.2.1.** Cold brand dyes preferred for heavy dark shades (DCT)**5.2.2.2.** Dye is pasted with Turkey Red Oil followed by dissolution in water and with little urea**5.2.2.3.** Fabric to be dyed is wetted in bath or loaded in jigger**5.2.2.4.** Exhaustion is carried with only addition of dye for 30 minute with half dye solution in first turn and rest half in the second turn at 30°C**5.2.2.5.** Then Sodium chloride is added 30-50gram/liter and dyeing continued till temperature 40-45°C, for 1-2 hour**5.2.2.6.** 5-10 gram/litre Soda ash--Na₂CO₃ added and fixation will continue for 45 min**5.2.2.7.** Drop bath**5.2.2.8.** Check the shade and adjust, if necessary**5.2.2.9.** Cold rinse 10 minute**5.2.2.10.** Drop bath**5.2.2.11.** Hot rinse at 65°C 10 minute**5.2.2.12.** Drop bath**5.2.2.13.** Set soaping bath at 38°C with 1 gram/litre anionic surfactant and 1 gram/litre soda ash, Run for 15 minute at 100°C**5.2.2.14.** Cool to 70°C**5.2.2.15.** Drop bath**5.2.2.16.** Give hot rinse 65°C 10 minute**5.2.2.17.** Give cold rinse 10 minute**5.2.2.18.** Drop bath and finish

Salt and alkali for cold brand reactive dyes should be as in the table 4 below:-

Shade %	COMMON salt (g/l)	Soda ash at M : L Ratio (g/l)		
		1:15	1:20	Jigger
Below 0.5%	25	5	3	5
0.5% - 2.0%	35	5	4	10
2.0% - 4.0%	45	10	8	15
4.0% & Above	55	15	10	20

5.2.3. Dyeing of woven fabrics with Remazol dyes or Mixed Bifunctional (ME) in jigger in (all-in method)

- 5.2.3.1. Dyeing is started at 40-45°C with a half dye solution, along with half salt and half alkali for the first turn
- 5.2.3.2. Followed by rest half of dye, salt and alkali in the second turn
- 5.2.3.3. Dyeing is continued for 1 hour and temp raised to 80°C
- 5.2.3.4. After which 45 min is further followed for fixation
- 5.2.3.5. Check the shade and adjust, if necessary
- 5.2.3.6. Washing and soaping as usual.

Salt and alkali for vinyl sulphone and Mixed Bifunctional (ME) type reactive dyes should be as in the table 5 below:-

Shade %	Glauber's salt (g/l)	Soda ash (g/l)
0.05%	3-5	10-15
0.05% - 0.1%	3-8	10-15
0.1% - 0.5%	15-25	10-20
0.5% - 1.0%	25-35	20
1.0% - 2.0%	40-50	20
>2.0%	50	20

5.3. Process control in batch dyeing

5.3.1. Winch dyeing

- 5.3.1.1. Temperature of dyeing: Generally for cold brand dye—30-40°C and for hot brand dyes 60-80°C.
- 5.3.1.2. Dye bath pH: For cold brand reactive dyes (high reactive dyes), milder pH -- pH 8.8-10.0 is used. For hot brand reactive dyes stronger pH-- pH 10.5 to 11.0 is used
- 5.3.1.3. Time of dyeing: Exhaustion time is 20-30 minute is used. Increase beyond this has no effect on exhaustion. For fixation 30- 60 min used depending on the depth of shade
- 5.3.1.4. Endless ropes or loops of fabrics to be loaded should be equal length (about 50-100m).
- 5.3.1.5. As for all forms of rope dyeing, the fabric should be fairly resistant to length ways creasing.
- 5.3.1.6. The maximum motion speed of the fabric should be approximately 40m/min

5.3.2. Jigger dyeing

- 5.3.2.1. Batching: Knife cut batching (very even batching). If batching is not even, then segregate the fabric or dry the fabric on the stenter. During batching another precaution is that the expanders of the jigger must work properly.

- 5.3.2.2.** MLR: Jigger dyeing machine is 1:1 to 1:1.5. During colouration 1:1.5, during soaping 1:2, during soaping for overflow washing it is 1:5. Maintain constant water level for over flow washing.
- 5.3.2.3.** Prevent water and steam leakages.
- 5.3.2.4.** Crease free fabric: Creases are found near the selvages. So it has to be removed immediately. Creases are formed in between the fabric due to the improper working of expanders.
- 5.3.2.5.** Colour and Chemical Addition: Importantly colour and chemical addition should be in even installments and not in odd installments.
- 5.3.2.6.** Proper stirring of the solution in the trough after every addition. All solid chemicals must be added by pre-dissolving.
- 5.3.2.7.** Temperature: During dyeing with the help of the thermometer check the temperature and the standard of temperature should be as per the class of the dye.
- 5.3.2.8.** pH: Check pH during dyeing as per the class of the dye. Give optimum dose of alkali and acids in the trough.
- 5.3.2.9.** Addition of colours for the correction of the shade
- 5.3.2.10.** Speed should optimum and constant (avoid speed above 80metee/min)
- 5.3.2.11.** Ensure that addition must be done in the jigger by draining half quantity of trough solution. The reason is that the new dye requirement to set the shade right will be less as the dye initially present has reduced to half the quantity plus in addition there will be shift in equilibrium and hence dyeing will again start and lead to shade correction.

6. SEMI CONTINUOUS DYEING PRINCIPLES, STANDARD OPERATING PROCEDURE AND PROCESS CONTROL

6.1. Standard operating procedure for cold pad batch

6.1.1. All pad-batch processes should follow the following general sequence:

- a) Impregnation of the well-prepared dry fabric in a solution of dye and alkali at ambient temperature.
- b) Uniform squeezing of surplus liquor from the fabric as it passes through the mangle nip.
- c) Wrapping of the batched roll of wet fabric in polythene film and storage at ambient temperature for a specified dwell time (2–24 h, depending on dye reactivity and pH).
- d) Washing-off of unfixed dye.
- e) Drying of the washed dyeing.

6.1.2. This should be used for dyeing of dark shade.

6.1.3. For pad-batch dyeing, reactive dyes with relatively high reactivity should be used. Some examples are Dichlorotriazine, Monofluorotriazine, Dichloroquinoxaline, Vinyl sulfones and selected Bi-reactive dyes.

6.1.4. Auxiliary chemicals in the pad-batch dye liquor should include alkalis, urea, and wetting agents.

6.1.5. A wetting agent or surfactant which is low foaming and which provides rapid wetting and penetration of dye into the cotton fiber should be added to the dye liquor.

6.1.6. The recipe for cold pad batch should be based on the table below:

Table 6: Sample cotton reactive dyeing recipe on cold pad batch

PH: 11			
Temperature: RT (padding & dwelling)			
Speed: 6-60m/min (depends on the fabric weight) (for canvas 25m/min, light weight fabrics 50m/min)			
Batch dwelling time: 8-24hr (varies with reactivity of dyestuff used)			
DPF (dye pick up factor):-average 70%			
Inputs (dyes/ chemicals/ auxiliaries)	Deep shade	Medium shade	Light shade
Dye solution			
Reactive dye [% On weight of fabric]	x	y	z
Urea [g per L of solution]	50	30-50	20-30
Sequestering agent [ml per L of solution]	2	2	2
Wetting agent [g per L of solution]	2-4	2-4	2-4
Alkali solution			
Caustic soda [ml per L of solution]	8	4	0.6
Soda ash [g per L of solution]	20-30	20-30	20-30
Ratio of application (Alkali: Dyestuff soln.)	1:4	1:4	1:4

6.2. Process control cold pad batch dyeing

- 6.2.1.** When preparing the pad-batch dye bath, the dyestuff and alkali should first be prepared in separate tanks as separate solutions.
- 6.2.2.** Automatic mixing of the dyestuff and alkali solutions should be a ratio of 4 to 1. This mix feeds into a small, 5 gallon (19 liter) holding tank for uniform blending prior to transfer to the pad trough.
- 6.2.3.** The dyestuff solution should be prepared by dissolving the dyestuff completely in water at the temperature recommended by the dyestuff supplier.
- 6.2.4.** To avoid evaporation from the exposed surfaces and edges of the roll, the fabric is stitched to a somewhat wider end-cloth that is padded and finally wrapped around the entire wet roll before it is covered with a plastic film
- 6.2.5.** The required alkali solution should be prepared in a separate tank and the temperature kept close to the padding conditions.
- 6.2.6.** Metering pumps made by several companies (listed under suppliers) should be used to provide adequate and reproducible mixing of pad-batch dye baths over extended periods of use.
- 6.2.7.** Final bath temperature after mixing should be kept between 18-300C to ensure consistency and reproducibility.
- 6.2.8.** Before padding the dye solution onto the fabric the following should be checked:-
- a) Fabric style and accurate weight of dye lot.
 - b) All dispensing lines must be clean.
 - c) Pad trough must be rinsed out well.
 - d) Check function of mixing pump blending ratio of alkali and fiber-reactive dye.
 - e) Check pad pressure side to side and center.
 - f) Check wet pickup.
 - g) Calculate volume of formula based on wet pickup to have sufficient dye liquor for the entire lot.
 - h) Check all functions on operating panel.
- 6.2.9.** Although high-reactivity dyes may be preferred because they enable dwell times as short as 2–4 h to be used, demands for specific hues, fastness or cost advantages may favour applying low-reactivity dyes at high pH for longer dwell times, possibly 16–24 h or longer
- 6.2.10.** Dyes having similar affinities should be used to avoid tailing.
- 6.2.11.** Running conditions that allow a complete change of liquor in the trough every 2–3 min throughout the run are ideal for most purposes.
- 6.2.12.** Padding should be done at uniform wet pick-up across the width.
- 6.2.13.** Mangle pressures are set to give a liquor pick-up of 60–80% at ambient temperature.

6.2.14. In pad–batch dyeing with Remazol dyes, the buffering action of sodium meta-silicate added to the NaOH solution is claimed to virtually eliminate the problem of paler selvages from carbon dioxide absorption.

7. CONTINUOUS DYEING PRINCIPLES, STANDARD OPERATING PROCEDURE AND PROCESS CONTROL

7.1. General principles of continuous dyeing

7.1.1. The objective of padding should be to impregnate the fabric with a solution or dispersion of dyes and chemicals as uniformly as possible, both lengthways and side-to-side.

7.1.2. To produce long lengths of fabric, all with the same colour, requires the maintenance of constant padding conditions over long periods of time

7.1.3. The fabric should first pass through the pad trough, where it is saturated with dye solution, and then between the rollers, where the pressure squeezes air out of the fabric and the solution into it.

7.1.4. Continuous dyeing processes should follow these four stages:-

- i. Dye application by padding;
- ii. Dye fixation, usually in hot air or steam;
- iii. Washing-off of unfixed dye and auxiliary chemicals;
- iv. Drying, usually on steam-heated cylinders.

7.2. Standard operation procedure for dyeing of woven fabrics with continuous processes

7.2.1. Pad—steam should be:-

7.2.1.1. Recipe X g/l dye, 10-20 g/l soda ash, 1g/l of wetting agent (penetrating agent), salt 60-90 g/l, resist salt 2g/l.

7.2.1.2. The application is padding---steaming (90 sec)—washing

7.2.1.3. The method should be used for high quality pale shades giving high rate of productivity.

7.2.1.4. High expression padding mangles should be used.

7.2.1.5. Cold brand reactive dyes should be used.

7.2.2. Pad-dry-pad (alkali)-steam (PDPS dyeing procedure should be:-

7.2.2.1. Padding(dye liquor)

- 4 gm/litre dye (Vinyl Sulphone dye)
- 50 gm/litre Urea
- 2 gm/litre Sodium Alginate
- 50 gm/litre common salt
- Pick up 60-80%

7.2.2.2. Drying

- Infrared drying (after infra red drying moisture content is 30%)

- Hot flue drying 120-130°C
- After drying, the fabric should be cooled on cold cylinders to avoid heating the chemical pad solution.

7.2.2.3. Padding(chemical liquor)

- 200-250 gm/litre common salt (Reduces the bleeding of dye from padded material to alkaline bath)
- 15ml/litre Caustic soda 360Be
- 20 gm/litre soda ash

7.2.2.4. Steaming

- 30-60 second with saturated steam

7.2.2.5. Washing off

- Through washing

7.2.3. Pad (Dye + bicarbonate+ urea) - dry should be:

7.2.3.1. This should be used valuable process for cheap and easy production of pale and medium depth of shade. (Realization of colour is poor)

7.2.3.2. This should be used valuable process for cheap and easy production of pale and medium depth of shade. (Realization of colour is poor)

7.2.3.3. Should use slow and even drying in hot flue drier to avoid migration and uneven dyeing.

7.2.3.4. Sodium bicarbonate should be used as a fixation alkali and hot flue dryer with free moisture are two important parameters for better fixation.

7.2.3.5. Minimum moisture for this process is around 20%.

7.2.3.6. The dye solution also contains a high concentration of urea (100 g l⁻¹). This helps the cotton fibres retain water during drying, possibly provides a fluid medium for dye diffusion in the fibres at low water contents, and increases dye solubility.

7.2.3.7. The pad-dry process is only suitable for reactive dyes with fairly high reactivity. For dyes of lower reactivity, the dried fabric must be baked to promote further fixation.

7.2.4. Pad (Dye + bicarbonate+ urea) – dry-bake should be:

7.2.4.1. The pad-dry-bake process is suitable for less reactive dyes such as MCT (Monochlorotriazines) dyes.

7.2.4.2. Typically for pad –dry-bake, the dyes are padded with

- 10–20 g l⁻¹ sodium bi carbonate and
- 100–200 g l⁻¹ urea.

7.2.4.3. Little fixation takes place during intermediate drying (Typical conditions are 2–5 minute at 105 °C in a hot flue or on heated cylinders at 105 °C. The fabric temperature will not exceed 70 °C)

7.2.4.4. And a baking treatment (e.g. 3 minute at 160°C) is necessary to complete the dye–fibre reaction.

7.3. Process control in continuous dyeing

7.3.1. General issues

- 7.3.1.1. Marks on the padded fabric from un-dissolved dye particles or un-dispersed pigment, or those arising from foam picked up from the surface of the dye liquor in the pad, which is squeezed onto the fabric surface should be avoided.
- 7.3.1.2. Suitable wetting and deaerating agent, low foaming type should be used for uniform padding and rapid wetting with dyestuff solution.
- 7.3.1.3. Suitable anti-migrating agent should be used to avoid variation due to migration during intermediate drying.
- 7.3.1.4. Dyestuff used for padding should be of medium to low affinity to avoid tailing.
- 7.3.1.5. Padding trough should be jacketed and be maintained at required temperature by circulating cold water to avoid increase in temperature during padding due to heat entrapped in fabric, batched in roll form after drying on cylinder.
- 7.3.1.6. All contact points must be cleaned during change over to avoid contamination.

7.3.2. Fabric-in-feed

- 7.3.2.1. The fabric should be free of creases and have flat selvages. It is accurately centered on the rollers and runs under moderate tension.
- 7.3.2.2. Open width fabric should be processed under uniform tension, before entering the padder and during take-up on the beam.
- 7.3.2.3. Speed and temperature control should be there.
- 7.3.2.4. Synchronized fabric guidance - in feed / take up should be used.

7.3.3. Nip rolls

- 7.3.3.1. Nip rolls should be clean
- 7.3.3.2. The covering of both bowls should be of uniform hardness to prevent two sided problems
- 7.3.3.3. Damage to the nip rolls should be maintained to avoid poor appearance of the dyed fabric and pick-up variations.

7.3.4. Trough content

- 7.3.4.1. The volume in the pad trough should be relatively small (20 – 50 litres) and automatically controlled at a constant level throughout the run.(Pad bath level variations result in a shade differences caused by variations in immersion time & pick up
- 7.3.4.2. The pad liquor turnover time should be as fast as possible
- 7.3.4.3. The pad liquor should be distributed uniformly across the full width of the padder through a perforated tube

7.3.5. Pick up

7.3.5.1. Should be uniform across the width and consistent from batch to batch (60 – 80% for cotton fabrics)

7.3.5.2. Factors which Pick-up depends upon should be controlled.--Pad pressure and speed (fabric penetration, wetting out), dye level in the trough (dip to nip time), dye bath chemicals (penetration accelerant), fabric (fibre and construction).

7.3.6. Immersion time

7.3.6.1. Should be 1 – 2 s for cotton

7.3.7. Speed

7.3.7.1. should be constant throughout the run - No stoppages

7.3.8. Padding temperature

7.3.8.1. Pad bath temperature is critical and should be within the specification.

7.3.9. Migration

7.3.9.1. To control dye migration, the following factors should be kept— Lower pick-up, Skying between padding and drying, Infra-red pre-drying (most important), Well balanced air flow in the dryer, Use of a migration inhibitor and Better dye selection Migration stops when the moisture content of the fabric has dropped below 30%

7.3.10. Infrared pre-drying

7.3.10.1. Temperature/capacity adjusted depending upon weight of fabric, Moisture content of fabric after pre-drying and Cleaning of guide rollers

7.3.11. Hot flue Driers

7.3.11.1. Uniform air distribution, Temperature control, Uniform fabric tension to prevent creasing, Graduated temperature between the hot flue chambers, Drying time – no over drying and Cooling down of fabric after drying

7.3.12. Cylinder driers should be

7.3.12.1. Sufficient steam pressure and supply should be there--Prevent condensation in the cylinders, Cylinders should be properly aligned, well balanced and clean and Fabric cooling after drying should be used

7.3.13. Steaming

7.3.13.1. Steam Fixation requirements should be:-

- a) Maintain control for an air free atmosphere of saturated steam, Sufficient steam pressure and supply
- b) Measurement of wet & dry bulb temperatures
- c) Prevention of condensation drops (heating of steamer entrance & roof)
- d) Fabric cooling before padding
- e) Control chemical pad recipe and pick up

- f) Fabric tension and steaming time(20 -60 sec)
- g) A steamer typically holds about 30–60 m of fabric. Again, good roller alignment is necessary to avoid lengthways creases.
- h) The roof of the steamer should be well insulated.
- i) The saturated steam in the box should be slightly superheated (105 °C).
- j) Water seal

8. QUALITY CONTROL IN DYEING

Through checking the shade of dyed fabrics, the fastness properties should be worked out. The quality of dyed fabrics is assured in laboratory, dyeing section and finishing section.

8.1. In laboratory:

- 8.1.1. Receive swatch card from buyers according to their requirement.
- 8.1.2. Dye sample by manually or CCMS.
- 8.1.3. Dye sample until matching with swatch card.
- 8.1.4. Fastness & other tests of fabrics can be also carried out.

8.2. In dyeing section

- 8.2.1. Dye sample in dyeing machine and dyeing shed, matching with the approved sample.
- 8.2.2. Move to bulk production, if the result is matched with approved sample.
- 8.2.3. During dyeing, take the samples when accurate shade matching accurate shade.
- 8.2.4. Collect dyeing sample which having been softened.
- 8.2.5. Last of all, collect sample from fixation & matched.
- 8.2.6. Allow fabrics to be finished.

8.3. DYEING DEFFECTS

The following dyeing defect should be checked:-

- 8.3.1. Shade variation
- 8.3.2. Batch to batch Shade consistency
- 8.3.3. Washing fastness, perspiration fastness, rubbing fastness, light fastness and Chlorine,
- 8.3.4. Dye Spots
- 8.3.5. Unevenness/unevenness
- 8.3.6. Contamination
- 8.3.7. White marks
- 8.3.8. Holes
- 8.3.9. Creases

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Organization and Objectives

The Ethiopian Standards Agency (ESA) is the national standards body of Ethiopia established in 2010 based on regulation No. 193/2010. ESA is established due to the restructuring of Quality and Standards Authority of Ethiopia (QSAE) which was established in 1998.

ESA's objectives are:-

- ❖ Develop Ethiopian standards and establish a system that enable to check whether goods and services are in compliance with the required standards,
- ❖ Facilitate the country's technology transfer through the use of standards,
- ❖ Develop national standards for local products and services so as to make them competitive in the international market.

Ethiopian Standards

The Ethiopian Standards are developed by national technical committees which are composed of different stakeholders consisting of educational institutions, research institutes, government organizations, certification, inspection, and testing organizations, regulatory bodies, consumer association etc. The requirements and/or recommendations contained in Ethiopian Standards are consensus based that reflects the interest of the TC representatives and also of comments received from the public and other sources. Ethiopian Standards are approved by the National Standardization Council and are kept under continuous review after publication and updated regularly to take account of latest scientific and technological changes. Orders for all Ethiopian Standards, International Standard and ASTM standards, including electronic versions, should be addressed to the Documentation and Publication Team at the Head office and Branch (Liaisons) offices. A catalogue of Ethiopian Standards is also available freely and can be accessed in from our website.

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International Involvement

ESA, representing Ethiopia, is a member of the International Organization for Standardization (ISO), and Codex Alimentarius Commission (CODEX). It also maintains close working relations with the international Electro-technical Commission (IEC) and American Society for Testing and Materials (ASTM). It is a founding member of the African Regional Organization for standardization (ARSO).

More Information?

Contact us at the following address.

The Head Office of ESA is at Addis Ababa.

☎ 011- 646 06 85, 011- 646 05 65

☎ 011-646 08 80

✉ 2310 Addis Ababa, Ethiopia

E-mail: info@ethiostandards.org,

Website: www.ethiostandards.org



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